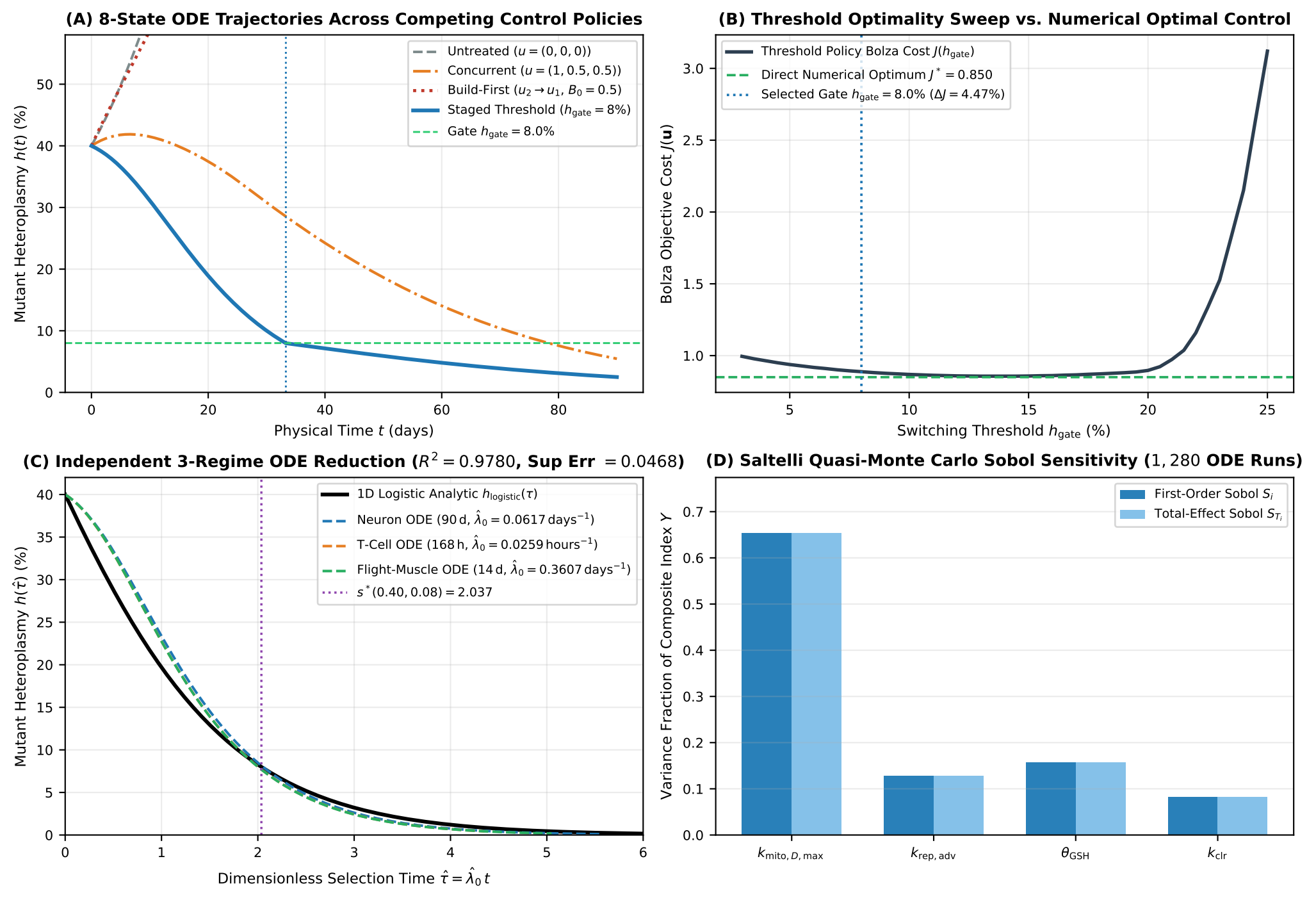
(A) 8-State ODE Trajectories Across Competing Control Policies
Untreated ($u = (0, 0, 0)$)
Concurrent ($u = (1, 0.5, 0.5)$)
Build-First ($u_2 \to u_1$, $B_0 = 0.5$)
Staged Threshold ($h_{\text{gate}} = 8\%$)
Gate $h_{\text{gate}} = 8.0\%$
Mutant Heteroplasmy $h(t)$ (%)
Physical Time t (days)
(B) Threshold Optimality Sweep vs. Numerical Optimal Control
Threshold Policy Bolza Cost $J(h_{\text{gate}})$
Direct Numerical Optimum $J^* = 0.850$
Selected Gate $h_{\text{gate}} = 8.0\%$ ($\Delta J = 4.47\%$)
Bolza Objective Cost $J(\mathbf{u})$
Switching Threshold h_{gate} (%)
(C) Independent 3-Regime ODE Reduction ($R^2 = 0.9780$, Sup Err $= 0.0468$)
1D Logistic Analytic $h_{\text{logistic}}(\tau)$
Neuron ODE (90 d, $\hat{\lambda}_0 = 0.0617\,\text{days}^{-1}$)
T-Cell ODE (168 h, $\hat{\lambda}_0 = 0.0259\,\text{hours}^{-1}$)
Flight-Muscle ODE (14 d, $\hat{\lambda}_0 = 0.3607\,\text{days}^{-1}$)
$s^*(0.40, 0.08) = 2.037$
Mutant Heteroplasmy $h(\hat{\tau})$ (%)
Dimensionless Selection Time $\hat{\tau} = \hat{\lambda}_0 t$
(D) Saltelli Quasi-Monte Carlo Sobol Sensitivity (1,280 ODE Runs)
First-Order Sobol S_i
Total-Effect Sobol S_{T_i}
Variance Fraction of Composite Index Y
$k_{\text{mito},D,\max}$
$k_{\text{rep,adv}}$
θ_{GSH}
k_{clr}